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COMP.SC.—2.3.5

PG 3rd Semester Examination, 2023 (R)

Subject : COMPUTER SCIENCE

(Advance Software Engineering)

Paper — 2.3.5

Full Marks : 80

Time : 3 hours

Answer all questions

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable

1. Preset a modular design for a "Library management system". State the functional requirements for this design. 20

Or

- (a) What are the different phases of software development life cycle ? Compare the

(Turn Over)

relative time and effort required to complete these phase.

10

- (b) What do you mean by software design ?
✓ Name the various system design methodologies and describe in brief the Data Design methodology.

10

2. (a) Compare the spiral software development life cycle model with the water fall life cycle model.

10

- (b) Prepare a software requirement specification document for a "banking system".

10

Or

- (a) Define cohesion. Outline coincidental cohesion, logical cohesion, temporal cohesion and functional cohesion with example.

10

- (b) What is a design-pattern ? Outline the model-view-controller architecture design pattern with an example.

10

3. (a) Define integration testing. Explain types of integration testing with example. 10
- (b) Discuss the software metric that can be applied to the qualitative assessment of software quality and side effects which occur during maintainance phase ? 10

Or

List and explain various cocomo cost estimation model. 20

4. (a) What is the difference between reverse engineering and reengineering ? Explain with example. 10
- (b) What is software maintainance ? Describe various catagories of maintainance and which catagories consume minimum effort and why ? 10

(4)

Or

- (a) What are the main requirements of ISO9000 ? How does this apply in software context ? 10
- (b) Explain how CMM helps continuous improvement in software process ? 10
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PG 3rd Semester Examination, 2023 (R)

Subject : COMPUTER SCIENCE

(Image Processing)

Paper — 2.3.4

Full Marks : 80

Time : 3 hours

Answer all questions

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable

1. Explain the concept of pixel in digital image representation. How does pixel intensity relate to the Colour and brightness of an image ? 20

Or

- ◀ (a) How histogram can be used to assess the contrast of an image ?

10

(Turn Over)

(b) Define histogram equalization. How does histogram processing contribute to image enhancement ? 10

2. Define quantization and explain its role in digital signal processing. How does quantization affect the representation of continuous signals in a digital form ? 20

Or

(a) Explain the concept of sampling in the context of digital signals. 10

(b) What is spatial filtering in image processing ? 10

3. Explain the importance of digital image processing. How does image compression contribute to efficient storage and transmission of images ? 20

Or

(a) Differentiate between Error Free compression and Lossless compression along with examples. 10

(b) Explain the concept of reversible transformations in the context of error-free compression. 10

4. Discuss the challenges associated with image segmentation. How does sementation contribute to the analysis and interpretation of images ? 20

Or

(a) What are the problems associated with thresholding in Image Processing ? 10

✓(b) Explain the significance of Motion Segmentation in computer vision and video analysis. 10